Project 2: Hardware Accelerator for CORDIC Algorithms

Introduction

The coordinate rotation digital computer (CORDIC) algorithm efficiently computes trigonometric, hyperbolic, and other transcendental functions using only shift, add, and subtract operations, eliminating the need for hardware multipliers. This makes it highly suitable for low-power or resource-constrained systems such as signal processing units, embedded devices, and FPGA-based accelerators. Hardware acceleration of CORDIC enables fast computation of sine, cosine, vector rotations, and magnitude calculations, which are widely used in communications, robotics, and control systems.

Project Description

In this project, you will design and implement a CORDIC hardware accelerator using SystemVerilog. You will implement the iterative algorithm for trigonometric and vectoring modes and explore design trade-offs between iteration depth, accuracy, and latency. You will optimize the architecture using pipelining or parallelism to increase throughput while minimizing resource usage. After verifying functionality through simulation, you will synthesize the design on an FPGA and measure its area, speed, and power efficiency.

Expected Results

You will deliver a fully functional CORDIC accelerator capable of computing trigonometric and vectoring operations efficiently. The hardware design should outperform a CPU-based software implementation in speed and energy efficiency, demonstrating the value of dedicated arithmetic accelerators for computationally intensive tasks.

Useful References

1. CORDIC Wikipedia
2. FPGA Implementation of CORDIC Algorithms for Sine and Cosine Generator
3. FPGA Implementation of the Trigonometric Function Using the CORDIC Algorithm